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Greenery from apartments: quantifying and comparing views with residents' perceptions

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Abstract

With urban densification and the proliferation of high-rise structures, residents' apartment views are getting obstructed from surrounding nature, especially greenery. Existing approaches often rely on simplified proxies or aggregated building- or floor-level metrics which does not capture individual-level variation. Most of them use coarse spatial data or subjective self-reports, lacking the granularity and precision to quantify the greenery visible to each resident from their own living space. This study introduces a conceptual and methodological framework for objectively modelling green views at the individual apartment level. Our Apartment Greenery View Measure was developed and assessed by (1) geocoding individual observer positions at window-level within apartment buildings, (2) implementing GIS-based three-dimensional viewshed analysis using high-resolution environmental datasets to objectively quantify views, and (3) examining agreement between modelled views and 445 residents' self-reported perceptions using the green-to-grey ratio. The method was applied to 30 apartment buildings across Melbourne, Australia. Findings reveal variability in green view exposure by building height, floor-level, and apartment orientation. A moderate correlation ($r=0.556$, $ICC=0.521$) shows the agreement between objective and perceived view measures, with 39.1% participants overestimating and 60.9% underestimating their views. This underscores the need for objective, standardised measures that move beyond perception alone. The workflow supports aggregation at multiple spatial scales, from individual units to floors and buildings, providing a flexible framework for assessing visual green equity citywide. This provides a scalable, low-cost tool for planners, designers, and health researchers seeking to integrate visual greenery into urban housing, policy, and equity-focused interventions.

Highlights

- Conceptualized an observer-centric green view measure from apartments.
- Streamlined viewshed analysis for precise quantification of observer-based views.
- Generated green view data for apartment buildings ($n=30$) across Melbourne.
- Objective green view measure was valid and aligned with resident perceptions ($n=445$).
- Demonstrates a cost-effective scalable method for generating city-wide green view data.

Keywords Green view index, Visibility analysis, Subjective environmental perception, Apartment window views, Semantic building model, Urban environment

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1 Introduction

With individuals spending up to 90% of their time indoors, whether due to remote work, lifestyle shifts, or urban living patterns (Kelly & Fussell, 2019; Wan et al., 2024), window view has become a crucial conduit to connect people with nature. The benefits of nature, especially greenery, extend beyond just physical access to parks or green infrastructures (Elsadek et al., 2020); instead increasingly shaped by what an individual can see from within their homes and workplaces, particularly in dense urban settings.

There is growing evidence of the positive impact of viewing urban greenery, such as street trees, parks and backyard plants, on human health and well-being (Jo et al., 2019; Meng & Wang, 2024). Research in environmental psychology, supported by theoretical frameworks like Attention Restoration Theory (Kaplan, 1995) and Stress Reduction Theory (Ulrich et al., 1991), and empirical studies underscore the restorative effects of viewing green from windows. These benefits include improved physical health (Mytton et al., 2012), enhanced work-life balance (van Esch et al., 2019), and positive mental well-being (Elsadek et al., 2020; Foster et al., 2024; Shin et al., 2020) fostered through relaxation, higher sleep quality, and reduced stress levels.

However, the role of greenery in supporting liveability is often overlooked in the context of contemporary urban development and high-rise building design (Lin et al., 2015). Compact city strategies, aimed at maximizing spatial efficiency and economic growth, frequently prioritize vertical expansion over open space preservation (Erlwein et al., 2023). This reduces availability of open spaces between buildings (Potter et al., 2023), resulting in densely packed urban layouts that often block views of nature, or replace them with grey elements such as concrete facades, steel structures, and paved surfaces.

This trend is also relevant in Australia, where metropolitan cities have transitioned from detached single-family homes to compact multi-storey apartment living (Foster et al., 2022). Existing urban greening policies emphasize physical access to greenspaces but overlook visual exposure from within dwellings (Hunter et al., 2019), potentially causing disparity in mental restoration and overall quality of life. Evidence from the *High Life Study* further supports the notion that Australian apartment design, orientation, and building placement can shape residents' environmental perceptions and well-being outcomes (Foster et al., 2022). As cities grow vertically, a shift toward individual visibility-based greenery metrics from apartment

buildings is essential to guide equitable planning and healthier apartment living.

However, quantifying such views can be spatially and computationally challenging, particularly in urban environments where building obstructions, complex view angles, and heterogeneous greenery distribution complicate measurement. To date, green views from buildings have been measured using subjective or objective approaches. Subjective methods use self-reported surveys, interviews, or on-site visual assessment to capture individuals' personal experiences of visible greenery (Ohly et al., 2016). While such methods offer individual-level data, they have limitations: (i) responses that can be influenced by individual preferences, cultural backgrounds, and personal experiences (Honey-Rosés & Zapata, 2023); (ii) use of simplified binary ratings ('yes' or 'no') failing to capture the complexity of the view (Leslie et al., 2010) and (iii) lack of spatial specificity, making it challenging to quantify the quality, or spatial distribution of greenery. These limitations hinder efforts to establish validity and translate findings into actionable planning and policy recommendations. Thus, much of the green view and health literature has acknowledged the shortcomings of relying solely on subjective measures, prompting the development and validation of objective approaches.

Objective methods, leveraging geospatial techniques, remote sensing data, and computational models, offer human-centered approach that better approximates how individuals view their surrounding greenery from buildings (I. Falfán et al., 2018a). For instance, Li et al. (2022) utilized panoramic imagery and deep learning to classify greenery, grey infrastructure, and water view components. While this approach produced detailed visual datasets, it falls short in capturing real-world observer variability, including differences in apartment layout, orientation, and surrounding context. In the context of geospatial modelling, Meng et al. (2020) and Wang et al. (2019) developed building- and floor-level visibility indices; however, their aggregated approaches overlook the individual-level nuances from specific apartment windows. Siyi Yu et al., (2016a) introduced a more advanced method using observer-based viewshed techniques yet lacked the precision to reflect unit-level or window-specific green view disparities. Only Bolte et al. (2024) addressed this gap by incorporating observer-specific parameters to assess visible vegetation from designated window viewpoints.

Despite advancements in objective measures, methodological gaps remain in existing studies, particularly the lack of validation and limited assessment of individual apartment-level variability, which is crucial in vertical housing where residents experience diverse visual

conditions. Another overlooked aspect is the lack of distance-sensitive weighting (i.e. distance decay function) in visibility assessments (Nutsford et al., 2015), where existing studies tend to treat visible greenery equally. Incorporating this function is crucial, as nearby greenery can exert stronger psychological and restorative impact than distant vegetation (Jiang et al., 2016), which often fades into the perceptual background.

Moreover, environmental psychology has long examined how people process information about the natural and built environments around them (Bell et al., 1996). Although individual, social, and cultural factors can influence these perceptions, it is reasonable to expect that objective and perceived environmental measures should align if they capture the same construct (Madison et al., 2009). However, past research has shown that there is often a mismatch between objective measures of greenness (such as NDVI or tree cover) and what residents actually perceive in their neighbourhoods (Leslie et al., 2010). As a result, evidence-based design and policy could be misinformed if they rely solely on objective measures without considering residents' perspectives. As most existing approaches lack individual-level modelling, understanding how objectively measured visibility translates to residents' actual perceptions is limited. Consequently, the extent of alignment or divergence between these measures, especially at the individual apartment level, remains poorly understood.

This study aims to advance understanding of apartment residents' views of surrounding environments, particularly the greening visible from their dwellings, by developing and assessing an objective green view measure. It seeks to explore: (i) differences in apartment views by built form, orientation, and elevation (ii) and evaluate alignment between objective green views and residents' environmental perceptions. To achieve this aim, the objectives are to:

- Establish a conceptual and methodological framework to model Apartment Greenery View Measure (AGVM), leveraging high-resolution spatial data and 3D building models.
- Develop an algorithm to quantify individual observers' apartment-level green (e.g., vegetation) and non-green elements (e.g., infrastructure, buildings, and surface water), enabling comprehensive profiling of visual exposure from each dwelling.
- Assess spatial variations in green and non-green views across building floors and apartment orientations.
- Evaluate objectively modelled measure with residents' self-reported environmental perceptions using

green-to-grey view ratios and assess perceptual bias and the extent of alignment.

2 Conceptual and methodological approach

An observer-centric approach is crucial for accurately capturing how residents perceive greenery from within apartments, as visual exposure is shaped by spatial factors such as their location (x, y, z), floor height, window orientation, and surrounding built and green structures. Generally, it is assumed upper-floor residents have broader green views due to fewer obstructions, while lower-floor views may be blocked by adjacent infrastructure. However, such patterns vary as visibility is not determined by height alone. Factors like building design, vegetation placement, and window orientation can significantly alter what a resident sees.

To capture this complex interplay of spatial elements within dense urban environments, the study employed Geographic Information Systems (GIS)-based *viewshed analysis*. It identifies the areas visible from a specified observation point by systematically accounting for variations in elevation, surface terrain, and three-dimensional obstructions such as buildings, trees, and other infrastructure (ESRI, 2024a) as illustrated in Fig. 1. This method enables quantification of the extent and composition of views, particularly available to observers from specific apartment building floors.

Building on the conceptual understanding, Fig. 2 presents the methodological workflow for developing Apartment Greenery View Measure by: (a) acquiring high-resolution environmental models and building footprints, (b) geocoding individual observers' positions (x, y, z) within 3D building models, (c) developing an algorithm to quantify each individual's view composition and (d) evaluating the objective measure with residents' self-reported environment perception to assess level of agreement. The following section outlines the methodology and implementation process in detail.

3 Implementation of objective Apartment Greenery View Measure

3.1 Study context

We utilised survey data from the *High Life Study*, a cross-sectional observational project designed to explore the influence of contemporary apartment design on residents' health and well-being across major Australian cities: Melbourne, Sydney, and Perth. Ethical approval for the study was granted by the RMIT University Design and Social Context College Human Ethics Advisory Network (21,146 & 25,922). The study is described in detail elsewhere (Foster et al., 2020).

The AGVM was implemented for *High Life Study* buildings in Melbourne, Australia. Buildings were

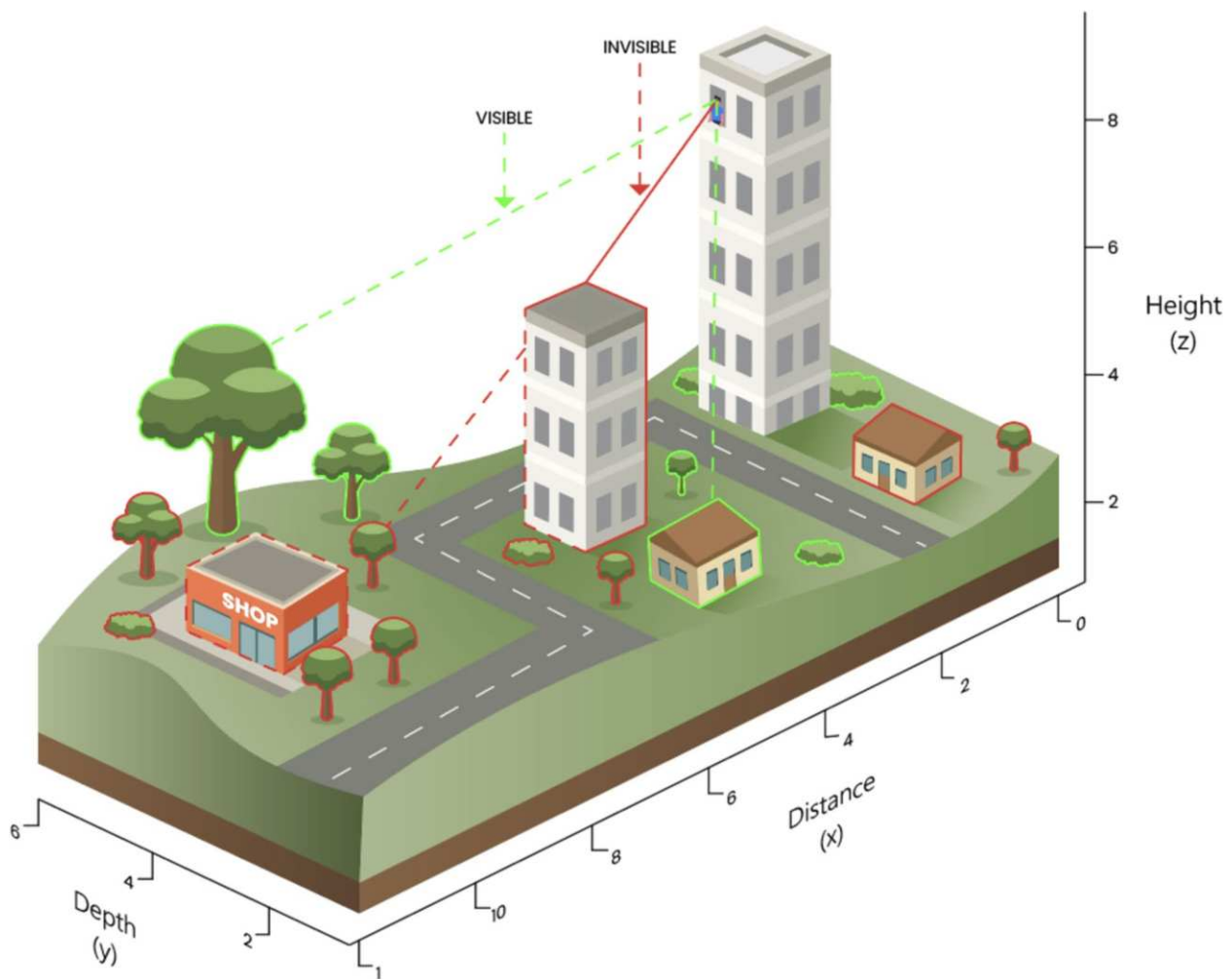


Fig. 1 Conceptual illustration of GIS-based viewshed analysis within a three-dimensional spatial environment showing how observer viewpoints, terrain, and urban obstacles define visible elements from apartment floors

located across the metropolitan area as shown in Fig. 3. Melbourne has seen a transition towards vertical urban growth, marked by substantial increase in high-rise residential developments and a proliferation of apartment complexes to meet rising population and housing demands (Chhetri et al., 2013; Foster et al., 2019). The city's heterogeneous built environment offers varied building typologies, architectural forms, and levels of green coverage, making it an ideal urban context for assessing residents' visual access to greenery from apartment buildings. In the methods and analyses, we distinguish between *observers* and *residents*. *Observer* refers to the modelled viewpoint locations used in the GIS-based visibility analysis (objective measures), whereas *resident* refers to survey participants who provided self-reported perceptions (subjective measures). This distinction was made to compare between modelled visibility and lived perception.

3.2 Data acquisition

3.2.1 Datasets for modelling GIS-based individual observer's views

High Life Study building data Building footprints ($n=30$) across Melbourne metropolitan area was obtained from *High Life Study* database (Foster et al., 2019). The *Study* adopted a sampling strategy to select buildings based on key criteria, including height, apartment counts, and construction year. In addition, buildings were distributed within different distances of the Central Business District (CBD), including inner city, inner urban, middle suburban, to outer suburban zones, to ensure variation in built environment conditions. Both building-level (e.g., height, roof type, form, and number of units per floor) and apartment-level attributes (e.g., floor number and main window orientation: north, south, east, or west) were also sourced from the *Study's* architectural plans.

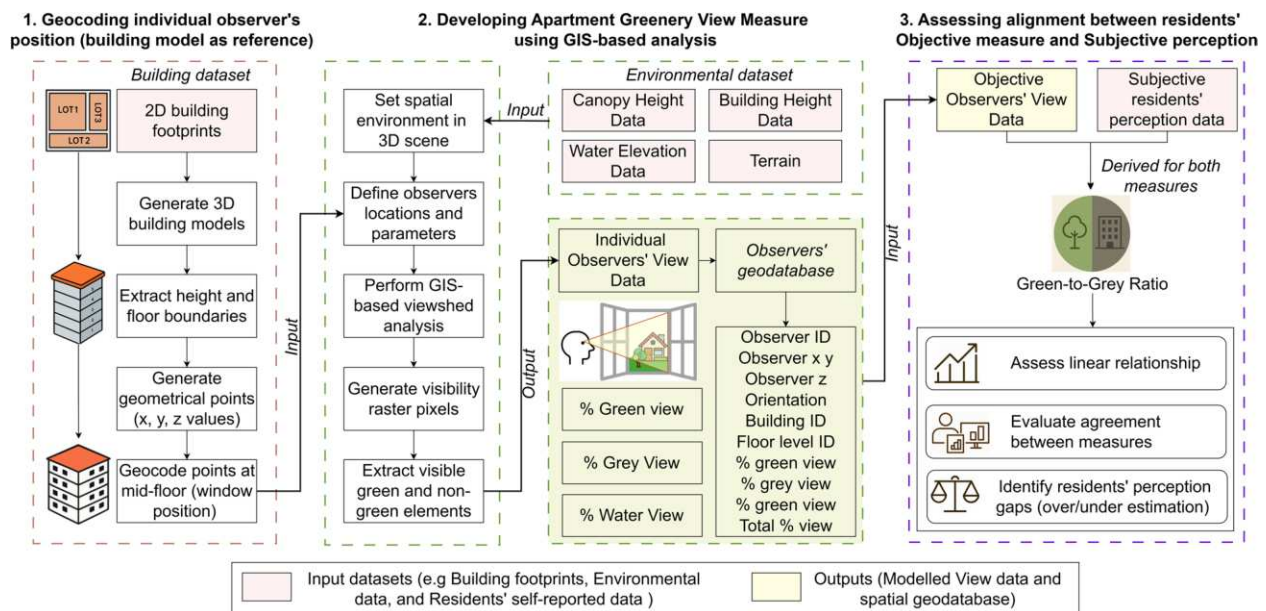


Fig. 2 Methodological workflow for developing and assessing the Apartment Greenery View Measure

Supplementary Table S1 summarizes the surveyed buildings by typology, floor range, and general location.

Raster-based high-resolution environment model We obtained high-resolution three-dimensional raster data models and aerial imagery of Melbourne from CSIRO Urban Monitor (UM) (Caccetta et al., 2012). The UM captured data in 2018 using stereo-aerial photography with a Microsoft UltraCAM-D camera, flown at 1300 m, with a ground sample distance of 30 cm for multispectral bands (red, green, blue, and near-infrared) and 10 cm for panchromatic data (Duncan et al., 2019; Hurley et al., 2019; Das et al., 2025). These overlapping images were processed using photogrammetric workflows (aerial triangulation, DSM generation, orthorectification) to produce ~20 cm high-resolution models. Independent evaluations have validated UM products against both LiDAR datasets and ground-surveyed GPS check points, reporting vertical accuracies with root mean square error typically <0.5 m (and ~0.17 m in open areas) and vegetation classification accuracies above 90%. Full details of the accuracy evaluation and processing pipeline are provided in Caccetta et al. (2012).

Using the UM high-resolution raster data model, we reclassified Canopy Height Model, i.e. vegetation heights into three types: grass (<0.5 m), shrubs (0.5–3 m), and trees (>3 m), providing detailed information on the quantity and structural complexity of greenery. Building Height Data model, showing the vertical morphology of the built environment, were distinguished into high infrastructure (multistorey buildings, towers, bridges)

and low infrastructure (single-storey buildings, garages, pavements, roads). The Surface Water Elevation Data model captured the extent and relative elevation of rivers, ponds, lakes, and wetlands. Together, these datasets, containing x, y, and z properties, were used to vertically represent the urban environment in our study.

3.2.2 High life residents' self-reported environmental perception

Residents' self-reported views ($n=445$) from their apartments were obtained from the *High Life Study* survey conducted in Melbourne in 2018. Participants were asked to estimate the percentage of different features they could see from their balcony (or living area window if they did not have a balcony). Response options included sky, nature (e.g., trees, parks and bushland), water (e.g., ocean, river, lake, wetland), concrete (e.g., roads, parking, paved areas) and buildings (e.g., housing, apartment blocks, shops and businesses). All items had high test-retest reliability (i.e., ICC: 0.846–0.966).

For alignment with AGVM, nature, water, concrete, and buildings variables were only selected to show participants' perception of natural and built elements. Baseline socio-demographic information was also collected which was categorised as: sex (*male and female*), age (18–34, 35–54, 54–75, and above 75), education level (*no formal education, completed primary school, completed secondary school, and completed bachelor's and higher*), household income (*\$0–\$60 k, \$60 k–\$100 k, \$100 k and above, not reported*), and time spent at home (*how many hours*

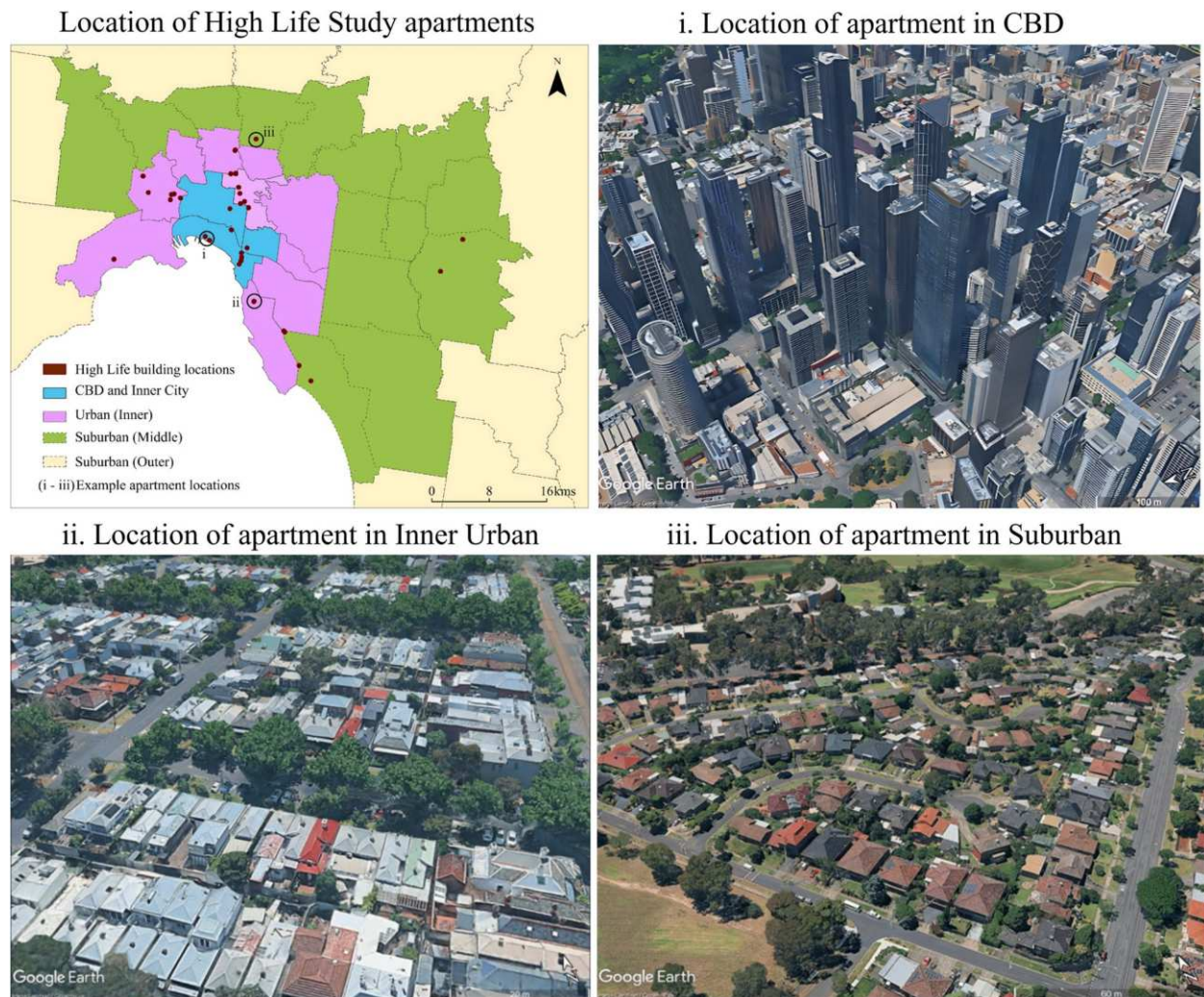


Fig. 3 Location of High Life Study building in Melbourne: (1) Central Business District, (2) Inner Urban area, and (3) Suburban area, illustrating the variation in surrounding built form and density across locations

per day do you spend at home on a usual Weekday and a usual Weekend between 8am and 8 pm?).

3.3 Geocoding individual observer's position within building models

Observer positions were geocoded using semantic 3D building models ($n=30$) as spatial reference. These models were generated in CityEngine software, a standalone procedural modelling platform, comprising of two core components: (i) Computer Generated Architecture Rules (CGA-rules) to encode building designs using parametric logic and procedural functions, and (ii) Level of Detail (LoD) standards, which defines the semantic and geometric granularity of the models ranging from LoD1 (basic extruded footprints) to LoD4 (detailed internal layouts and advanced architectural designs) (Biljecki et al., 2016; Kelly, 2021).

As illustrated in Fig. 4 (Step 1–2), a custom CGA script was developed and building attributes were linked (height, floor height, and roof type). The script extruded footprints to its corresponding height, segmented the structure into floor levels by applying a standard apartment floor height of 2.4 m in accordance with the National Construction Code (Architects, 2024; Australian Building Codes Board, 2019), and added uniform façade and roof structures for architectural detailing. This generated semantic LoD2 building models comprising 270 individual floors. Fig. 4 (Step 3) shows the semantic hierarchy of LoD2, encoding spatially explicit building elements, such as façade, roofs, and floors, enabling the models to function as accurate spatial references for positioning observer locations.

Observation points were then generated as geocoded proxies for residents' primary window-facing

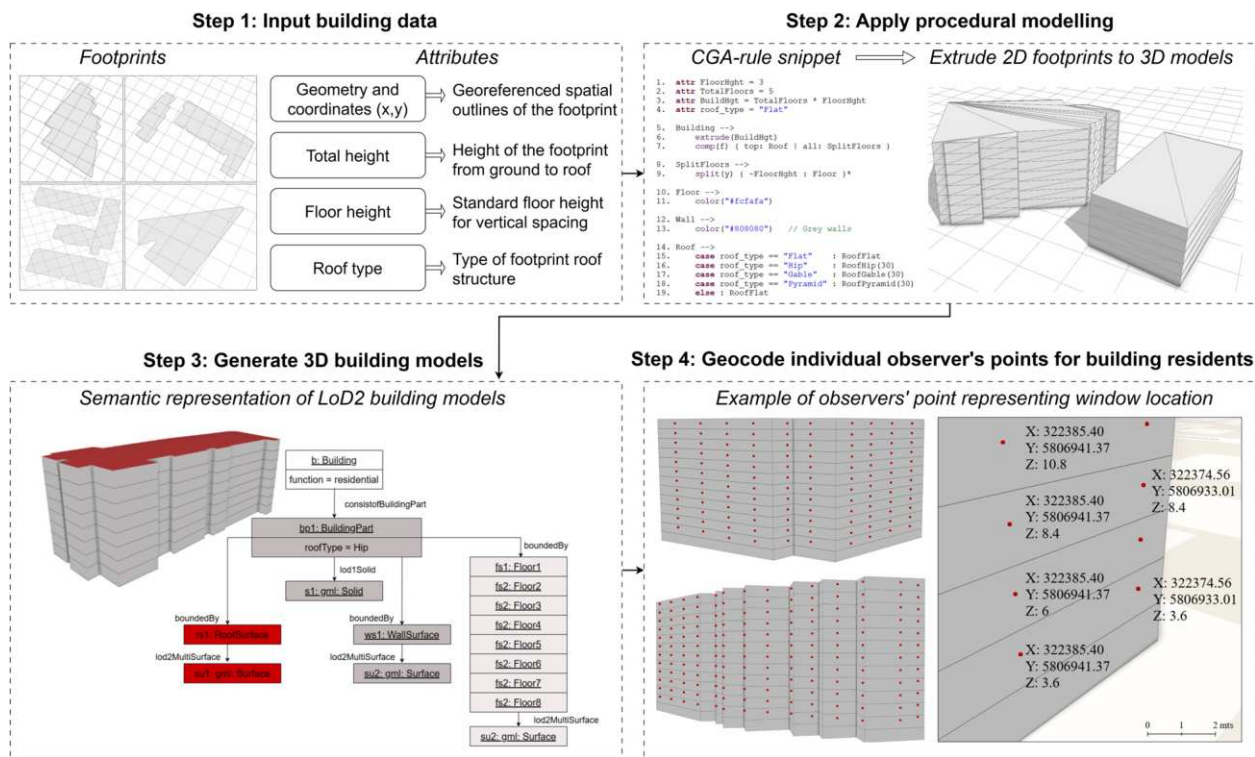


Fig. 4 Workflow for geocoding individual observer's position within 3D building models

viewpoints, approximated at the midpoint height of each floor, where a resident would typically stand to view outside from within an apartment. Points were spaced at 9-m intervals to approximate the distance between apartment windows, in accordance with the Apartment Design Standards for Victoria (Victorian Government, 2021). The process was automated using a custom toolbox script that calculated each floor's bounding geometry, generated a midline floor offset, positioned observation points along the midline at 9-m intervals, and assigned each point precise 3D coordinates (x, y, z) where x, y represents the georeferenced horizontal location, and z denotes the vertical height (Fig. 4 Step 4). Each observation point was assigned a set of building-level attributes such as building ID, floor number, and viewing orientation. Manual verification was performed by cross-referencing each observation point with 3D imagery and architectural apartment building layouts to exclude points located along non-visible facades, such as blank walls or areas without windows. This process generated 6,478 observation points, forming the spatial basis for observer-centric green view modelling.

3.4 Quantifying observers' green and non-green views in a three-dimensional environment

A spatially explicit three-dimensional environment was first constructed in local scene using high-resolution datasets, where DTM represented the terrain, Canopy Height Data defined vegetation structures, and Building Height Data and Surface Water Elevation Data characterized non-green elements.

An algorithm was then developed using GIS-based 3D viewshed spatial analyst library to calculate visual exposure from each observer point. The script required input parameters, including environmental layers, observer-specific attributes, and visibility principles. Observer parameters include spatial location (X, Y), eye-level height (Z), orientation (aspect/viewing direction), and field of view (FoV). As illustrated in Fig. 5, field of view refers to the angular extent of the vision by defining the directional horizontal and vertical boundaries within which visibility is analysed.

In the algorithm, the horizontal FoV (θ) was constrained to directional ranges relevant to window orientation (e.g., from 90° to 180° for), while vertical FoV (ϕ) captured the natural eye-level inclination from slightly downward to upward (e.g., -15° to +60°) to mimic a human eye:

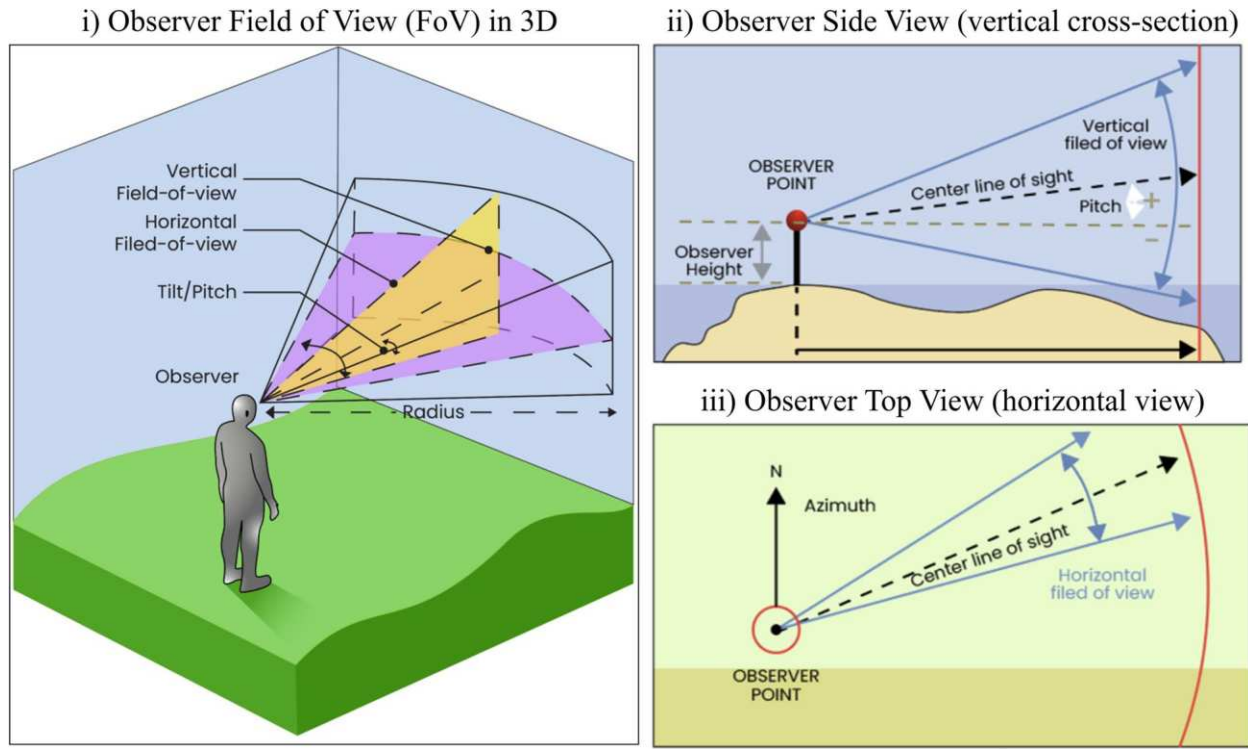


Fig. 5 Illustration of observer's field of view: (i) 3D observer cone with combined horizontal and vertical viewing angles, tilt, and maximum view radius; (ii) side profile showing vertical FoV; (iii) Top-down view depicting azimuth direction and horizontal FoV

$$\theta_{\text{horizontal}} \in [\text{Azimuth}_1, \text{Azimuth}_2], \varphi_{\text{vertical}} \in [\text{Vertical}_1, \text{Vertical}_2]$$

where Azimuth_1 and Azimuth_2 defines the horizontal angular limits, and $\text{Vertical}_1, \text{Vertical}_2$ specify the lower and upper bounds of the vertical viewing angle. A 100-m buffer was applied to limit the maximum visible distance, based on empirical evidence and its relevance to perceptual realism in urban visibility studies.

The viewshed algorithm uses Line-of-Sight (LoS) principle, internally computed using ray-tracing method. It casts imaginary straight lines from each observer's view-point to all target cells (e.g., land cover layers), determining visibility based on whether these lines intersect any obstructions (ESRI, 2024b; Fisher, 1996). If an obstruction is encountered, the target cell is marked as invisible; otherwise, it is considered visible. For any observer at (O_x, O_y, O_z) and a target cell at (T_x, T_y, T_z) , the parametric equation of LoS is defined as:

$$L(t) = (1 - t) \cdot (O_x, O_y, O_z) + t \cdot (T_x, T_y, T_z), t = [0, 1] \quad (1)$$

where, t is a parameter that interpolates between the observer ($t = 0$) and the target ($t = 1$), with intermediate values of t representing points along the LoS.

The coordinates at any point along the ray as:

$$x_t = (1 - t) \cdot O_x + t \cdot T_x \quad (2)$$

$$y_t = (1 - t) \cdot O_y + t \cdot T_y \quad (3)$$

$$z_t = (1 - t) \cdot O_z + t \cdot T_z \quad (4)$$

To determine visibility, the algorithm then compares the height of the LoS at each point t to the combined height of terrain and any obstruction at that location:

$$z_t > Z_{\text{terrain}}(x_t, y_t) + Z_{\text{obstruction}}(x_t, y_t) \quad (5)$$

where, z_t is the height of the LoS at the point t , $Z_{\text{terrain}}(x_t, y_t)$ is the elevation at (x_t, y_t) , and $Z_{\text{obstruction}}(x_t, y_t)$ is the height of any obstructions above the terrain.

If this condition is satisfied for all $t \in [0, 1]$, the target cell is marked as visible. This process generates a binary visibility raster layer, where each cell is assigned, a value based on its visibility status:

$$V_{(x,y)} = \begin{cases} 1, \\ 0, \end{cases} \quad (6)$$

where $V_i=1$ indicates the target cell is visible, and $V_i=0$ indicates target cell is invisible.

A map algebra expression was used to extract visible green pixels (grass, shrub, tree) by overlaying the binary visibility raster layer onto the Canopy Height Data, retaining only vegetation visible from each observer's location. In similar way, visible grey pixels (from Building Height Data) and blue pixels (from Surface Water Elevation Data) were extracted.

A distance decay function was then incorporated, assigning greater weight to nearby visible elements than diminishing visual influence of distant objects. For example, using a decay parameter $\beta=0.05$ (Inglis et al., 2022), a tree 10 m away has a weight of 0.61 compared to just 0.0067 at 100 m, highlighting how distant greenery contributes far less to perceived views. This was implemented in the algorithm through an exponential decay function based on Euclidean distance:

$$W(d) = e^{-\beta d} \quad (7)$$

where $W(d)$ is the visual weight, d is the the Euclidean distance, and β is the decay parameter controlling the rate at which visibility influence diminishes with distance.

Using this function, three composite view metrics were calculated to represent Apartment Greenery View Measure (AGVM):

$$\text{Observer \% Green View}_{ij} = \frac{\text{Weighted visible green area}_{ij}}{\text{Total weighted view able area}_{ij}} \times 100 \quad (8)$$

$$\text{Observer \% Grey View}_{ij} = \frac{\text{Weighted visible grey area}_{ij}}{\text{Total weighted view able area}_{ij}} \times 100 \quad (9)$$

$$\text{Observer \% View}_{ij} = \frac{\text{Visible surface water area}_{ij}}{\text{Total weighted view able area}_{ij}} \times 100 \quad (10)$$

where ij denotes the i th observer at j th apartment unit, weighted visible green area includes the distance-weighted sum of visible green pixels, weighted visible grey area includes distance-weighted sum of visible built-up infrastructure, weighted surface water includes distance-weighted sum of visible water features, and Total Weighted Viewable Area includes weighted visible pixel area within the observer's Field of View.

Figure 6 conceptually illustrates the observer-centric viewshed model in urban environment, showing how visible (green lines) and obstructed (red lines) greenery are determined by the observer's line-of-sight (solid and dashed black lines) within FoV. In addition to the composite view metrics, *percentages* for individual vegetation types (e.g., %_Tree View, %_Shrub View, %_Grass View) and infrastructure subtypes (e.g., %_High Infrastructure, %_Low Infrastructure) were also calculated separately.

Finally, a spatial geodatabase was developed for 6,478 geocoded observer points. Each point was annotated with building ID, floor level, spatial coordinates (X, Y, Z), aspect (viewing direction), and view percentages (both individual and composite metrics). This structure enabled granular analysis at multiple spatial scales, connecting individual views to overall building-level patterns.

3.5 Data analysis

3.5.1 Analysis of objective AGVM

Descriptive statistics were used to summarize green and non-green views across the buildings. Pearson correlation coefficients were calculated to assess the strength and direction of green and non-green view association for all observers ($n=6478$). A stratified analysis was conducted specifically for green views. Buildings were categorized into low-rise (1–3 floors), mid-rise (4–8 floors), and high-rise (9–17 floors) according to Australian Bureau of Statistics (Australian Bureau of Statistics, 2018). Observers' green view scores were then hierarchically assessed across building floor levels and apartment orientations to identify vertical and directional disparities in green view exposure.

3.5.2 Association between objective green view measure and subjective perceptions

To compare the AGVM, participants' self-reported perceptions and attributes were matched with objective observers' data using apartment and building information. We focused specifically on using Green-to-Grey Ratio (GGR) metric, ensuring comparability across both measures and representing the balance of green versus grey elements in each resident's view. Objective GGR was calculated as the ratio of % Total Green View to % Total Grey View, while Subjective GGR was computed as the ratio of self-reported % Nature View to % Grey View. This alignment allowed us to assess the extent to which modelled environmental exposures correspond with residents lived perceptions, and to identify potential sources of discrepancy (e.g., under- or overestimation of greenery).

The association were assessed using Pearson correlation coefficient to measure linear relationships, and Intraclass Correlation Coefficients (ICC) to evaluate agreement at two levels: (1) single-measure, comparing each participant's subjective GGR with their corresponding objective GGR, and (2) average-measure, comparing the mean subjective GGR across all participants with the mean objective GGR. Perceptual bias in GGR estimates was further conducted using an adjusted metric, following the standardized transformation approach adapted from Ina Falfán et al., (2018a). Taking the objective GGR as the benchmark, the larger of the subjective or objective value was divided by the smaller to calculate the

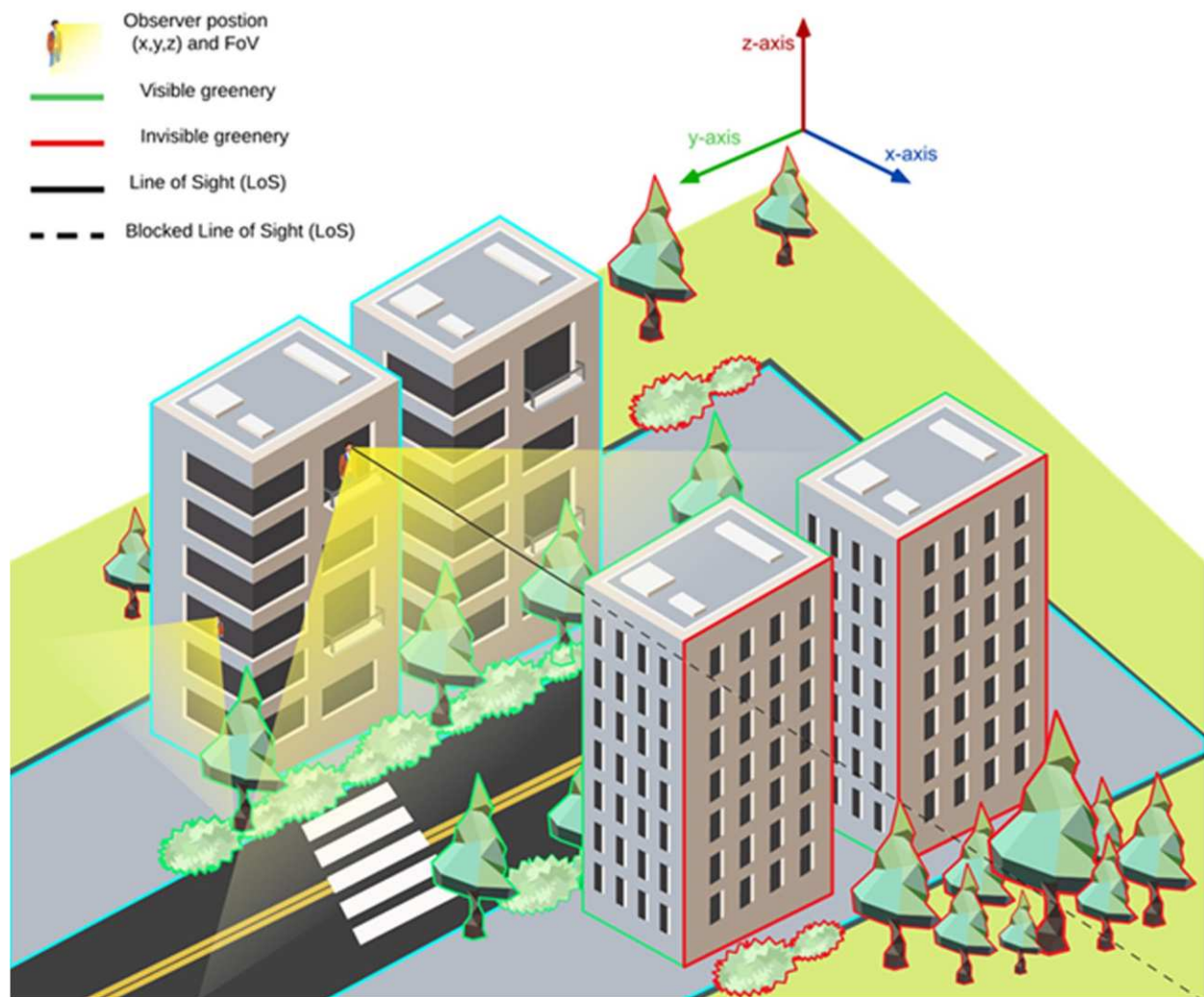


Fig. 6 Illustration of apartment-level green view assessment in urban environment

magnitude of discrepancy. To capture directionality, all values were then normalized by subtracting 1, centering perfect agreement at zero. In this scale, positive values indicated overestimation of greenery relative to grey perception, while negative values indicated underestimation.

4 Results and interpretations

4.1 Apartment Greenery View Measure summary across the buildings

Table 1 summarizes green and non-green view components for observers ($n=6,478$) across 30 buildings. Observer % Green view ranged from 6.85% to 99.77% with a mean of 23.12%. Individually, tree was a primary contributor (mean = 22.58%), while grass and shrub view contributed minimally. The Observer % Green view

Table 1 Summary of green, grey, and water view components for apartment observers, including individual and composite view metrics

Type of Views (individual and composite)	Minimum	Maximum	Mean	SD
% Grass	0	9.75	0.03	0.30
% Shrub	0.003	19.91	0.63	1.86
% Trees	6.85	98.90	22.58	17.11
Observer % Green View	6.85	99.77	23.12	17.05
% High Infrastructure	1.02	99.98	60.95	27.02
% Low Infrastructure	0.004	89.47	12.52	17.86
Observer % Grey View	1.13	100	73.48	23.16
Observer % Water View	0	79.59	4.94	15.80

ranged from 1.13% to 100, with a mean of 73.48%. Water views were limited, with a mean of 4.94%.

4.2 Individual-level variation in green and non-green views across buildings

The plots show the distribution of individual observers' green, grey, and water view percentages (Fig. 7). The x-axis represents the percentage of each view type, while the y-axis shows the density of observers within those ranges. Green view percentages were strongly left-skewed, with over 60% of residents experiencing less than 30% greenery, and most concentrated around the 10–15% range. In contrast, grey views were right-skewed, with majority of observers seeing more than 70% built infrastructure, largely driven by high-infrastructure elements. Water views were negligible, with over 90% of participants having less than 5% visibility, reflecting the buildings distance from surface water bodies.

Around 13.3% ($n=859$) of observers had low green visibility ($\leq 10\%$), while only 2.7% ($n=173$) had very high green views ($\geq 90\%$). In contrast, 25.8% ($n=1,672$) of observers experienced very high grey visibility ($\geq 90\%$), whereas only 2.4% ($n=153$) had low grey views ($\leq 10\%$), highlighting the dominance of grey views over greenery in observers' visual environment. These patterns are further supported by the correlation results presented in Table 2, where green views were negatively correlated with grey views.

4.3 Variation of observers' green views across building floors and aspects

Figure 8 presents mean Observer % Green View across buildings. Low-rise buildings exhibited green views ranging between 30–50%. Buildings 18, 19, 23, and 25 have 40–50% green views across all floors. Only building 17 displayed a $>70\%$ green compared to other low-rise buildings. Mid-rise buildings exhibited variability in green views, both across buildings and floor levels. For example, buildings 6 and 10 showed $>70\%$ green views on lower floors which declined to $<30\%$ on upper floors. Conversely, buildings such as 20, 24, and 27 had over 50% green views across all floors. Other buildings (e.g., 13, 15, 21) experienced $<30\%$ green views across their floor levels. High-rise buildings exhibited $<50\%$ green view percentages across most floors. Notable inter-floor variability was observed, particularly on the 9th floor of building 1 and the 6th and 18th floors of building 30, where green view values exceeded 50%.

The distribution of Observer % Green View across different apartment orientations is shown in Fig. 9. The plot combines three components: distribution (density) of green view percentages, boxplot summarizing the

median and interquartile range with outliers, and raw jittered points showing each individual observer's green view percentage. Median green view varied by apartment orientation: 22.7% for east-facing units, 22.0% for west-facing, 17.4% for northeast, 17.4% for northwest, 17.4% for north, 16.8% for southeast, 16.1% for south, and 14.9% for southwest, indicating east- and west-oriented apartments generally offered higher green view exposure. Presence of outliers reflects considerable variation in green view even among units facing the same direction.

4.4 Comparison of the observers' green views measure with residents' perceptions

4.4.1 Sample demographics

The *High Life Study* cohort ($n=445$) comprised of 67% female participants, with mean age range between 25–55 years. Most participants (69.5%) held bachelor's degree or higher, and 68.4% reported an annual income exceeding \$60,000. On average, participants spent 4.45 h ($SD=3.33$) at home on weekdays (8am – 8 pm), and 5.68 h ($SD=2.66$) on weekends, indicating in-home occupancy and visual interaction with their surroundings. In terms of dwelling characteristics, 60.0% of participants apartments were on lower floors (1st–3rd), 28.5% on mid-level floors (4th–8th), and 11.5% on higher floors (9th and above). The primary window orientation was distributed as 32.3% facing north, 25.6% west, 21.9% east, and 20.2% south.

4.4.2 Summary of residents' self-reported views

Table 3 presents descriptive statistics for participants self-reported view composition. On average, 12.86% ($SD=14.18$) of the view consisted of greenery. Water views were less common, with a mean of 1.75% ($SD=6.30$). In contrast, grey components (built-up areas) dominated the view, averaging 52.14% ($SD=24.05$).

The analysis showed that 67% of residents ($n=292$) had less than 10% green views, while only one reported over 90%. In contrast, 5.6% ($n=25$) of residents had over 90% grey views, compared to just 3.6% with under 10% grey view. Nearly all residents (96%) reported less than 10% water views, with none exceeding 90%.

4.4.3 Correlations and correspondence between objective and subjective green-to-grey ratio

There was a moderate positive correlation ($r=0.556$, $p<0.01$) between the objective and subjective green-to-grey ratio view measures (Table 4).

Findings from Intraclass Correlation Coefficient also indicated moderate agreement at the individual level ($ICC=0.521$) (see Table 5). This suggests that while individual participants' subjective assessments showed some

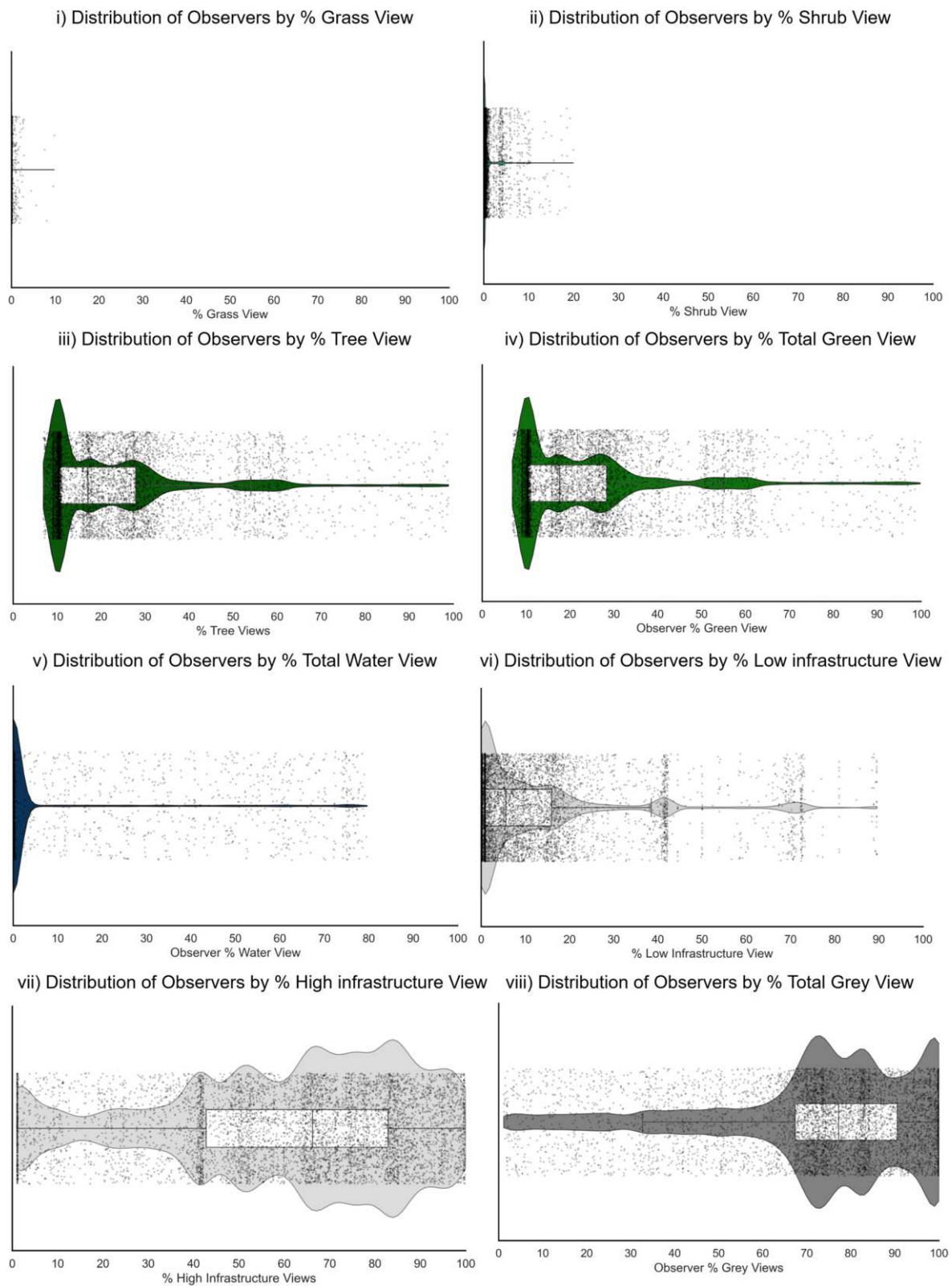


Fig. 7 Distribution of individual observers' view percentages across 30 apartment buildings

Table 2 Correlations between observers' green and non-green view

	% Grey View	% Water View
% Green View	-0.72*	0.19*

Correlation significant at 0.01 level (2-tailed)

alignment with objective measures, there were notable variations. At the group level, the agreement was stronger, with an Average Measures ICC=0.638, indicating good agreement between average subjective and objective green-to-grey ratios.

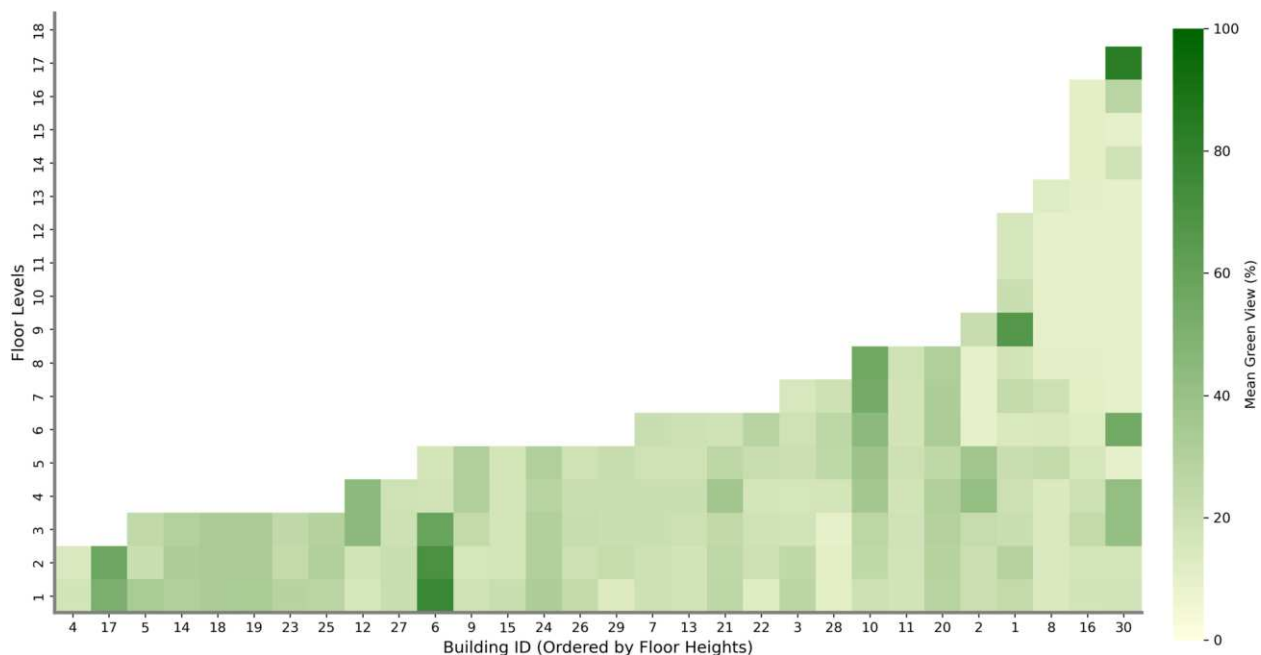
Building on the correlation and agreement analysis, the distribution of adjusted green-to-grey ratios provided further insights into individual-level discrepancies between residents' self-reported and objectively measured views (Fig. 10). Among the 445 participants, 39.1% ($n=172$) overestimated their green-to-grey ratio, reporting higher green views than objectively measured. In contrast, 60.9% ($n=268$) underestimated their green-to-grey ratio, perceiving lower green views relative to objective assessments. Density plots revealed that a majority of participants underestimated their green view compared to objective measures, clustering closely around minor negative bias values. In contrast, overestimations were less frequent but showed a wider variability, with some participants overreporting their perceived green views.

5 Discussion

5.1 Representing and modelling the spatial complexity of individual's views

We developed an Apartment Greenery View Measure to capture individual's views in dense urban settings by positioning observation points at window level and integrating high-resolution environmental datasets into a GIS-based viewshed analysis. Unlike prior studies that averaged green views at the building or floor level (Meng et al., 2020; S. Yu et al., 2016a), our observer-centric approach models views at the individual apartment level, capturing variability that aggregated indices overlook. Such individualised assessment is essential in compact urban environments, where differences in building height, apartment orientation, and surrounding obstructions can influence residents' green view experiences, even within the same building.

The use of high-resolution raster datasets allowed us to capture Melbourne's fine-scale environmental heterogeneity, enabling the stratification of vegetation into trees, shrubs, and grass, and built environment into high and low infrastructures. This disaggregation offers a detailed representation of environmental elements, often treated as a single category in previous studies (Bolte et al., 2024; Meng et al., 2020; Wang et al., 2019; S. Yu et al., 2016a), allowing for a closer approximation of what residents actually see from their apartments. Additionally, the application of a GIS-based framework provided the computational functionality and analytical capacity to conduct visibility analysis in 3D space (Wróżyński et al.,

**Fig. 8** Plot showing the Observer % Green View across buildings and floor levels

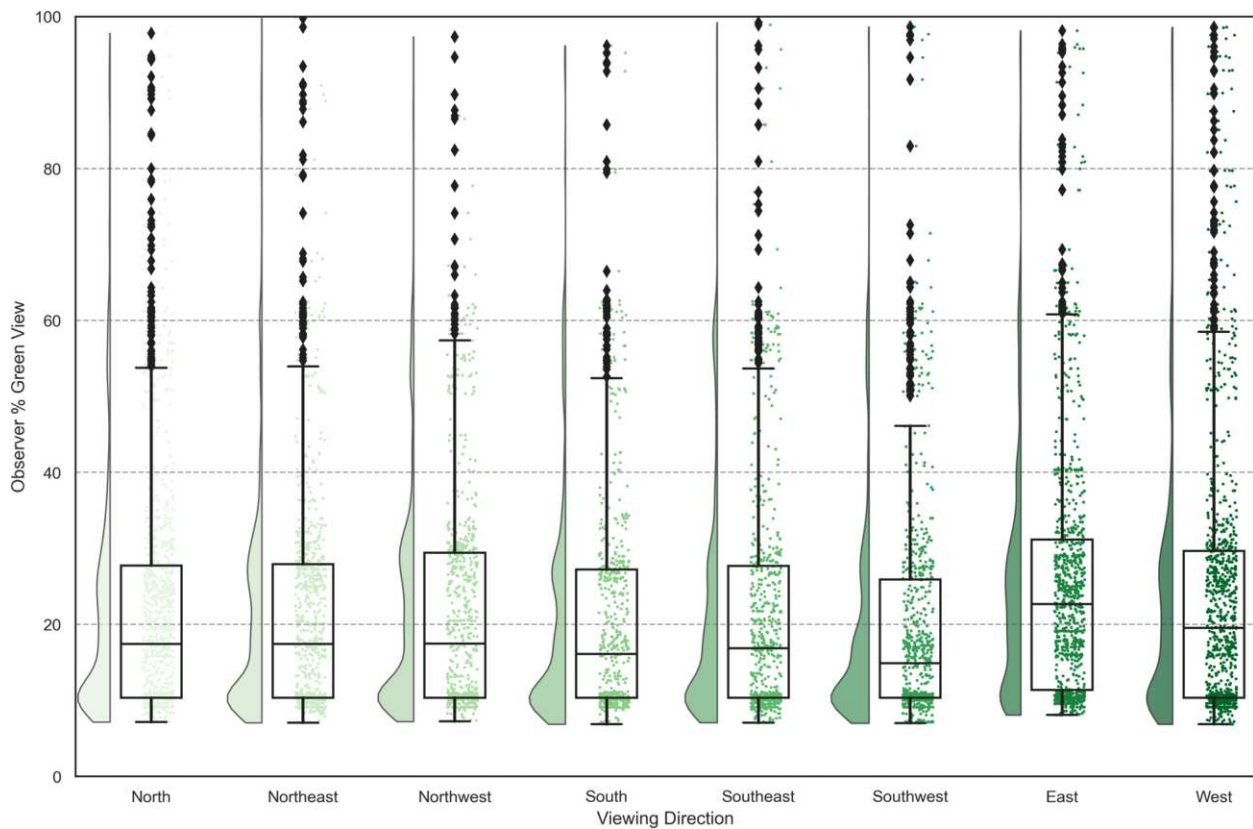


Fig. 9 Distribution of Observer % Green View across viewing orientations

Table 3 Summary of resident's self-reported views ($n = 445$)

Types of view	Minimum	Maximum	Mean	Std. Deviation
% Green View	0	100	12.86	14.18
% Water View	0	80	1.75	6.30
% Grey View	0	100	52.14	24.05

Table 4 Pearson correlation coefficient between Objective and Subjective Green-to-Grey Ratio

Measure	Subjective Green-to-Grey
Objective Green-to-Grey	.556**

** Correlation significant at 0.01 level (2 tailed)

2024), enabling the simulation of observer-specific views that account for variations in observer height, terrain elevation, built obstructions, and vegetation structure.

However, capturing structural visibility alone does not fully represent how people perceive their environment. In reality, human eyes do not see all visible objects equally, closer elements dominate attention, while distant features recede into the background. Traditional viewshed analyses rely on binary classifications (visible = 1, not visible = 0) which oversimplify how views are experienced (Cimburova & Blumentrath, 2022; Nutsford et al., 2015). Similarly previous visibility studies have not incorporated this perceptual weighting, treating all visible greenery as equal regardless of distance (Bolte et al., 2024; Wang et al., 2019; S. Yu et al., 2016a). To address this limitation, distance decay function was incorporated to shift from

Table 5 Intraclass Correlation Coefficient (ICC) for subjective and objective green-to-grey ratio

Measure	ICC Type	ICC Values	95% Confidence Interval (CI)	Agreement level	Significance (p-values)
Green-to-grey ratio	Single	.521	.46-.597	Moderate	<.001
	Average	.638	.485-.646	Good	<.001

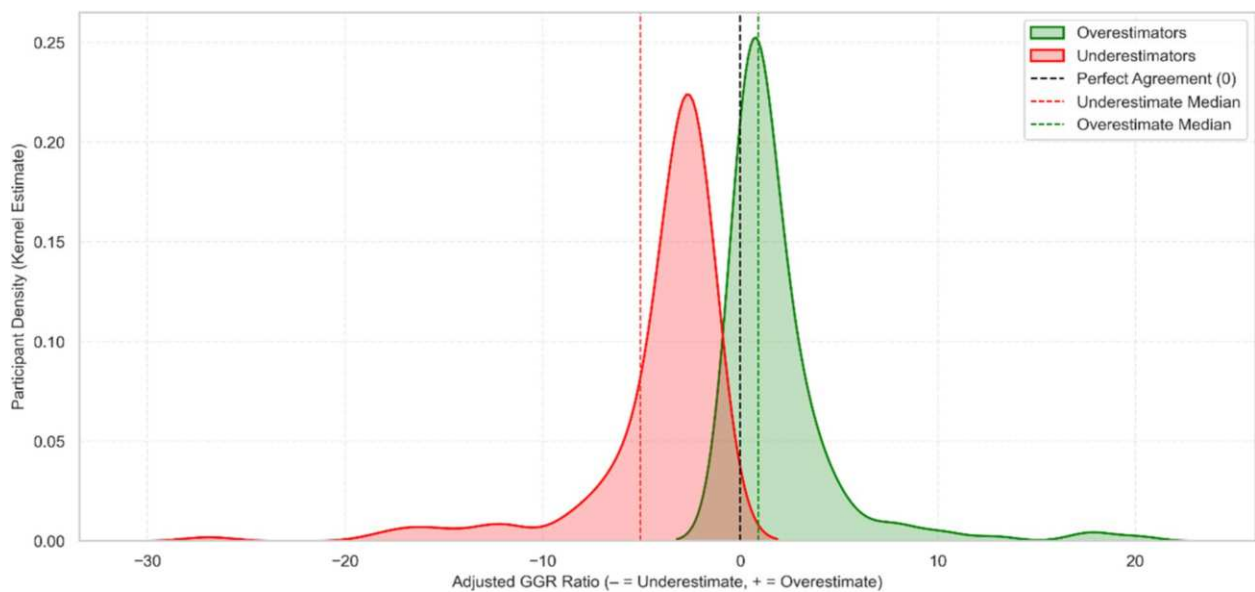


Fig. 10 Distribution of adjusted Green-to-Grey Ratios (GGR) by estimation type

‘what’s theoretically visible’ to ‘what’s actually seen. Studies in environmental psychology emphasize that proximity enhances the restorative value of natural views, as nearby greenery exerts a stronger psychological impact than distant vegetation (Chiang et al., 2017; Kaplan & Kaplan, 1989). Therefore, by assigning greater weight to nearby vegetation, the measure more accurately reflects individuals’ view of their surroundings.

We applied this measure across 30 buildings in Melbourne to assess patterns of individual observers’ green view across buildings, floor levels, and orientations. This multi-scalar perspective revealed intra- and inter-building green view variation. While it is often assumed that higher floors provide better green views, our findings demonstrate that this relationship is nonlinear. Units on mid-floors (typically between the 4th–8th floors) sometimes offered higher visibility due to proximal tree canopies, while upper floors despite seeing a broader area experience ‘visibility decay’ from distant vegetation. Orientation also played a significant role, where apartments facing open spaces demonstrated greater green view proportions than those facing denser built-up areas.

Beyond green view assessment, the AGVM also quantified views of grey (built infrastructure) and water elements, offering a holistic depiction of residents’ visual environments. Although water visibility was minimal in this study, largely due to the buildings being located away from major surface water features, the model demonstrated the capacity to incorporate such elements where relevant. Importantly, a negative correlation was observed between green and grey view proportions,

revealing a spatial trade-off within constrained urban settings. These findings underscore how spatial arrangement, vegetation density, and building form co-determine the composition of what residents see, reinforcing the need for visibility-based metrics that can differentiate not only *how much* but also *what* elements of the environment are visible. Thus, AGVM represents a methodological advancement over earlier green view indices, achieving greater computational efficiency for application in dense urban contexts.

5.2 Validation of objective green view measure using subjective data

An important contribution of this study lies in assessing the objective green view measure with residents’ subjective perceptions to evaluate the accuracy of modelled green views. While the objective measure quantifies views, residents’ subjective perception reflects their cognitive and affective interpretations of surrounding greenery shaped not only by what is physically visible but also by prior experiences, expectations, cultural background, and psychological factors. As previous studies have noted (Ina Falfán et al., 2018a; Leslie et al., 2010), these measures reflect different dimensions of views: the measurable versus the experienced.

Studies that have directly compared objective and subjective assessments of greenery, such as those using street-level indices (e.g., Green View Index) or neighbourhood-scale metrics (e.g., NDVI or buffer analyses), have often reported weak or inconsistent alignment between the two (Larkin & Hystad, 2019; Leslie et al.,

2010). This misalignment is due to coarse spatial granularity and generalised coverage of these indices, which assess greenness exposure across broad areas rather than an individual's view. In contrast, our study employed the Green-to-Grey Ratio (GGR), a metric that captures the proportion of natural versus built elements within each resident's actual field of view. As AGVM captures the environment as visually experienced, not just geographically mapped, we found a moderate alignment with residents' subjective perceptions. Most residents (61%) underestimated their green views compared to the objective measure. This discrepancy likely arises because the objective measure, based on high-resolution 3D visibility modelling, captured fine-grained spatial details, such as partial canopy views, peripheral greenery, and distant vegetation, that residents may not consciously notice or recall. Moreover, subjective perceptions are influenced by psychological and contextual factors (Kaplan & Kaplan, 1989), including low environmental satisfaction or limited visual engagement, leading individuals to overlook greenery that is technically visible.

This analysis affirms that objective and subjective green view measures capture distinct but complementary realities. Neither can be considered more accurate than the other, as each reflects a different facet of reality: objective measures quantify the physical presence and visibility of green elements, while subjective measures reveal how individuals perceive, interpret, and emotionally respond to those views. Combining both is crucial for gaining a holistic understanding of visual exposure to greenery and in informing equitable, health-supportive urban design interventions that are both evidence-based and human-centered.

5.3 Relevance to urban planning and design practice

The AGVM provides practical value for urban planning, design, and health research by quantifying how greenery is seen from individual dwellings. Unlike traditional greening metrics that focus on horizontal access, such as park proximity or canopy cover per capita (De La Iglesia Martinez & Labib, 2023; Li et al., 2025), AGVM accounts for vertical and directional variations across floor levels, building orientations. This highlights the need for visibility-sensitive planning approaches, such as setting minimum green view thresholds in apartment design policies or prioritising greening interventions where residents currently lack visual exposure to greenery. It can support planning as a diagnostic tool during the design review process to evaluate whether proposed buildings deliver adequate views to greenery, while for existing buildings it could guide retrofitting and greening upgrades around facades or shared spaces.

Our measure also has potential for the evaluation of the “3–30–300” rule (Konijnendijk, 2023), which advocates

that every urban resident should (1) see at least three trees from their home, (2) live in a neighbourhood with at least 30% tree canopy cover, and (3) be within 300 m of a public green space. While this rule is increasingly referenced in city-level greening strategies, the first component, visibility from home, has the least supporting evidence (Li et al., 2024). AGVM offers a scalable way to operationalize, test and monitor this component, ensuring the rule's implementation is not just spatially accurate but perceptually relevant. In particular, our measure has applications in population health research. Exposure to urban greenery has been associated with stress reduction, improved sleep, and enhanced well-being. However, most studies rely on self-report view exposures or coarse, areal measures of greenness. AGVM offers a more precise alternative. Capturing individual-level views will assist in linking to individual's health and well-being outcomes, supporting more targeted evaluations of urban design and greening interventions. Thus, our method can both improve the rigor of green view and health research by providing more accurate exposure measures, and support real-world planning practice by linking greening guidelines to measurable design and policy standards.

5.4 Limitations and future research

This study presents a scalable and practical framework for modelling apartment-based green views, but limitations merit discussion. Our framework prioritizes computational efficiency by using simplified 3D urban datasets rather than LiDAR-derived or BIM (Building Information Modelling) inputs. The measure is also a static viewshed model, providing a foundational snapshot of visibility patterns. This approach does not account for dynamic elements such as seasonal variation in canopy cover or daily sunlight and shadow conditions, both of which may alter the quantity and quality of greenery visible from apartments. However, it establishes critical baseline metrics that future temporal studies can build upon. Incorporating seasonal or diurnal changes in vegetation visibility could enhance our understanding of when and how green views matter most across time.

In addition, the sky component was not considered in the objective AGVM, although participants reported the percentage of sky in their self-assessments. This is because it represents open background space rather than a quantifiable environmental feature. Future studies could incorporate sky to explore its role in perceived openness, restorative quality, or satisfaction with apartment views.

While this study focused on visibility modelling and alignment with residents' subjective perceptions, the output of the AGVM can also serve as an exposure metric in health-related research. Specifically, future research could link AGVM-derived exposures to mental health outcomes

such as stress, psychological distress, or well-being indices, allowing for stronger dose–response evidence to inform policy. Moreover, behavioural data, such as usage patterns or time spent indoors, can be integrated to contextualise how visible greenery is actually experienced.

In summary, the current model strikes a practical balance between analytical precision and computational feasibility. It offers a flexible foundation for future enhancements, whether for refining the spatial realism of urban visibility modelling or for exploring its relevance to human health and environmental justice.

6 Conclusion

This study introduced an observer-centric measure that effectively captured variability in green views across different building heights, orientations, and urban contexts. By integrating high-resolution spatial data with objective assessment techniques, the measure showed differences in green visibility influenced by elevation, window orientation, and surrounding urban morphology. The approach demonstrated reliability by aligning moderately with subjective perceptions at the individual level and showing stronger agreement when aggregated, reinforcing its accuracy in representing real-world visual experiences. While subjective perceptions varied, reflecting factors such as personal preferences, cognitive biases, and environmental context, the observer-centric measure offers a standardized, scalable, and low-cost method for assessing green view accessibility without requiring detailed building data.

These findings highlight the value of the AGVM as a scalable and cost-effective tool for quantifying visual exposure to green, grey, and blue elements across diverse urban settings. By enabling high-resolution assessments at the individual apartment level, the measure addresses key limitations of prior health research that relied solely on subjective reporting. It also provides a practical means of evaluating compliance with emerging urban greening benchmarks such as the "3–30–300" rule, thereby offering actionable insights for both research and urban design aimed at promoting equitable access to greenery in high-density environments.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43762-025-00220-x>.

Supplementary Material 1.

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Authors' contributions

Shinjita Das: Conceptualization, Methodology, Writing – review & editing, Writing – original draft, Algorithm development, Data curation, Validation and Analysis. Chayn Sun: Conceptualization, Methodology, Validation, Reviewing, Editing, and Supervision. Alan Both: Data curation and Reviewing. Sarah Foster: Writing – review & editing, and Supervision, and Funding Acquisition.

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Data availability

Not available to share. The data are not publicly accessible due to ethics restrictions e.g. their containing information that could compromise the privacy of research participants.

Declarations

Competing interests

No conflict of interest was reported by the author(s).

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